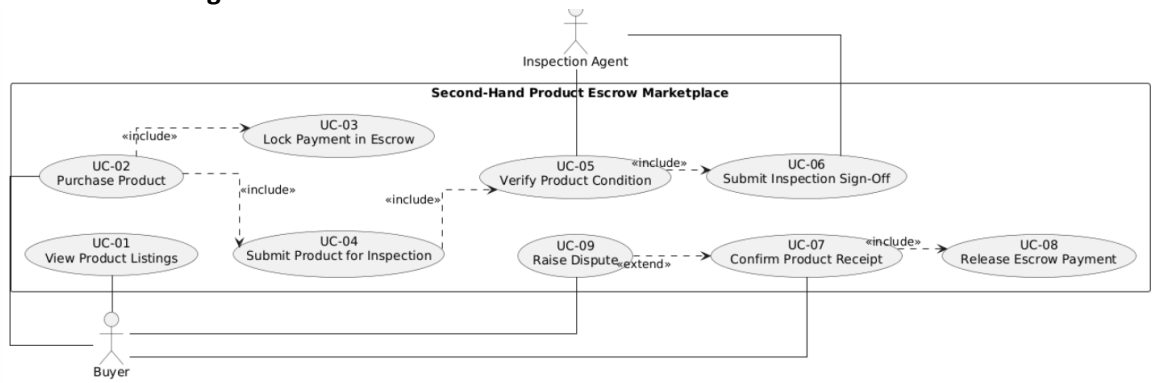
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LAB 1

Use case flow diagram



Requirements table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall lock buyer payment in escrow and transition order state to 'Under Inspection' until an inspector submits physical verification sign-off.	High	Pass: funds held securely and released only on dual verification. Fail: funds released prematurely to seller.	Protects buyer funds until product condition is independently verified — the core trust mechanism of the marketplace.
FR-002	Functional	The system shall allow an Inspection Agent to submit a physical verification sign-off recording the product's condition against the seller's listed description.	High	When the Inspection Agent submits a sign-off with a condition rating and remarks, order state changes from "Under Inspection" to "Inspection Complete" within 5 seconds; a submitted sign-off cannot be edited.	Creates an independent, tamper-resistant record required for dual verification before funds release.
FR-003	Functional	The system shall allow the buyer to review the inspection report and provide a final buyer sign-off (completing	High	When the buyer submits sign-off after the inspection report exists, order state changes to "Verified" and funds status changes to	Dual verification (inspector + buyer) is the exact release condition

		dual verification) before escrow funds are released.		"Released to Seller" within 10 seconds; buyer sign-off is rejected if no inspection report exists yet.	specified in FR-001.
FR-004	Functional	The system shall allow a seller to list a second-hand product for sale with title, description, condition claim, and price.	High	When a seller submits a listing with all required fields, the product appears in the catalog with status “Available” within 5 seconds.	Foundational step that supplies the product the Inspection Agent will later verify against.
FR-005	Functional	The system shall allow the buyer or seller to raise a dispute ticket at any point before funds are released.	Medium	When either party selects “Raise Dispute,” a dispute ticket is created and the order state changes to **“**Disputed” within 2 seconds	Gives both parties a way to contest the inspection outcome, triggering the freeze behavior specified in NFR-001.
NFR-001	Nonfunctional—Performance & Security	Dispute resolution tickets must freeze escrow funds immediately and prevent account withdrawals during active investigation.	High	Pass: benchmarking confirms funds freeze within 2 seconds of ticket creation, and no withdrawal API call succeeds against frozen funds, verified under simulated peak load.	Prevents either party from moving disputed funds while a claim is unresolved.
NFR-002	Nonfunctional—Security	The system shall ensure inspection sign-offs are immutable and verifiable once submitted, so condition records cannot be tampered with after the fact.	High	A security audit confirms 100% of sampled sign-off records pass integrity verification, and any attempted post-submission edit is rejected and logged.	The buyer's release decision depends entirely on trusting the inspector's report.

Use-Case Flow: Purchase Product / Escrow Transaction

Actors: Buyer (primary), Inspection Agent

Preconditions:

- Buyer is registered and logged in.
- Product is available for purchase.
- Buyer has a valid payment method.

Postconditions:

- **Success:** Inspection is completed, buyer provides final sign-off, escrow funds are released, and the transaction is completed.
- **Failure:** If a dispute is raised, escrow funds remain frozen and the transaction stays in **"Disputed"** state until resolution.

Main Success Scenario

1. Buyer selects an available product and initiates a purchase.
2. System locks the buyer's payment in escrow and changes the order state to "Under Inspection."
3. System submits the product for independent inspection and notifies the Inspection Agent.
4. Inspection Agent physically verifies the product against the seller's listed description.
5. Inspection Agent submits a verification sign-off with condition rating and remarks.
6. System changes the order state to "Inspection Complete**"** and makes the inspection report available to the buyer.
7. Buyer reviews the inspection report and provides final buyer sign-off.
8. System confirms that inspection sign-off and buyer sign-off are complete, releases the escrow funds, and changes the order state to "Verified."
9. System sends confirmation to the relevant parties and closes the transaction.

Alternate Flow — 7a. Buyer Raises a Dispute

7a1. Instead of providing sign-off, the buyer selects "Raise Dispute."

7a2. System immediately changes the escrow status to **"**Disputed – Held" and prevents fund release.

7a3. System creates a dispute ticket and prevents withdrawals against the frozen funds during investigation.

7a4. An authorized reviewer records the dispute resolution decision.

7a5. System either releases the funds to the seller or refunds the buyer and closes the transaction accordingly.